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Pixel array package structure and display panel

Abstract

A pixel array package structure includes: a substrate; a pixel array disposed on the substrate, in which the pixel array includes a plurality of light emitting diode chips, and the light emitting diode chips include at least one red diode chip, at least one green diode chip, at least one blue diode chip, and a combination thereof; a reflective layer disposed on the substrate and between any two adjacent of the light emitting diode chips; a light-absorbing layer disposed on the reflective layer and surrounding the pixel array; and a light-transmitting layer disposed on the pixel array, the reflective layer, and the light-absorbing layer, in which the light-transmitting layer has an upper surface and a lower surface opposite thereto, and the lower surface is in contact with the pixel array, and the upper surface has a roughness of 0.005 mm to 0.1 mm.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. application Ser. No. 17/663,431, filed May 15, 2022, which is a Divisional Application of the U.S. application Ser. No. 16/232,041, filed Dec. 25, 2018, now U.S. Pat. No. 11,367,849, issued Jun. 21, 2022, which claims priority to Taiwan Application Serial Number 107129351, filed Aug. 22, 2018, all of which are herein incorporated by their entireties.

BACKGROUND

Field of Invention

(1) The present disclosure relates to a pixel array package structure and a display panel.

Description of Related Art

(2) With rapid advancement of display technology, in addition to a conventional liquid crystal display panel, a display panel uses a RGB light emitting diode (RGB LED) as a display pixel and a light source. Since the LED has the advantage of low power consumption, the display panel using the LED can effectively save the power required by the conventional liquid crystal panel, which can help energy saving, carbon reduction and environmental protection, and thus is expected to become a new potential product in the display technology field. The RGB display cannot achieve small pitch operations due to process limitations of surface mounted devices packaging processes. However, current displays move toward development of chip on board packaging processes, such that sub-millimeter LEDs (mini LEDs) and micro LEDs can achieve small pitch operations.

(3) In the technique of the three primary color display, in addition to the color vividness, there is an index that determines the sharpness of the screen, that is, the contrast. In the manner of increasing the contrast in a small pitch, in addition to using a black plate as the substrate, light-absorbing

particles are added into the encapsulant. The light-absorbing particles can not only reduce the influence of the lateral light of the LED but also absorb the light source from outside to increase the contrast. However, it causes the problem of reduced light output of the forward light.

(4) In addition, the current large-sized display panel splicing technology is to stack a plurality of small display panels with each other to form a large-sized display panel. Since there are gaps at the frame of the display panel and between the displays, there is a problem that the screen is cut and not continuous when the large-sized display panel displays, which affects the quality of the display. In order to improve the above problems, a display panel with a narrow frame has been developed to narrow the gaps between the displays during splicing. However, even if the display panel with the current narrowest frame is used, a certain safety distance should be retained to avoid damage caused by the mutual pressing of the display panels, so that the large-sized display panel still has splicing gaps.

SUMMARY

(5) In view of the above, a purpose of the present disclosure is to provide a pixel array package structure and a display panel that can solve the above problems.

(6) To achieve the above purpose, an aspect of the present disclosure provides a pixel array package structure including a substrate, a pixel array, a reflective layer, a light-absorbing layer, and a light-transmitting layer. The pixel array is disposed on the substrate. The pixel array includes a plurality of light emitting diode chips. The light emitting diode chips include a red diode chip, a green diode chip, a blue diode chip, and a combination thereof. The reflective layer is disposed on the substrate and between any two adjacent of the light emitting diode chips. The light-absorbing layer is embedded in the reflective layer and surrounds the pixel array. The light-transmitting layer is disposed on the pixel array, the reflective layer, and the light-absorbing layer. The light-transmitting layer has an upper surface and a lower surface opposite thereto. The lower surface is in contact with the pixel array.

(7) A further aspect of the present disclosure provides a display panel including a plurality of sub-display panels. Any two adjacent of the sub-display panels has a splicing gap, in which each of the sub-display panels includes a plurality of the pixel array package structures described above.

(8) It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the present disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

(2) FIG. 1A is a microscopic top view of a pixel array package structure according to one embodiment of the present disclosure;

(3) FIG. 1B is a partially enlarged schematic view of a region R1 of FIG. 1A;

(4) FIG. 1C and FIG. 1D are schematic cross-sectional views taken along line A-A' and line B-B' of FIG. 1B, respectively;

(5) FIGS. 2, 3, 4, and 5 are top views of pixel array package structures according to various embodiments of the present disclosure;

(6) FIG. 6A and FIG. 6B are schematic cross-sectional views taken along line A-A' and line B-B' of FIG. 1B according to another embodiment of the present disclosure;

(7) FIG. 7A and FIG. 7B are schematic cross-sectional views taken along line A-A' and line B-B' of FIG. 1B according to a further embodiment of the present disclosure;

(8) FIG. 8 is a top view of a pixel array package structure according to a variant embodiment of the

present disclosure;

(9) FIG. 9A is a microscopic top view of a pixel array package structure according to one embodiment of the present disclosure;

(10) FIG. 9B is a partially enlarged view of a region R2 of FIG. 9A;

(11) FIG. 9C is a schematic cross-sectional view taken along line A-A' of FIG. 9B;

(12) FIG. 10A is a top view of a pixel array package structure according to one embodiment of the present disclosure;

(13) FIG. 10B is a schematic cross-sectional view taken along line A-A' of FIG. 10A,

(14) FIG. 11 is a graph comparing brightness of pixel array structures of the present disclosure with a conventional pixel array structure;

(15) FIG. 12 is a top view of a display panel according to one embodiment of the present disclosure; and

(16) FIG. 13 is a partially enlarged view of FIG. 12.

DETAILED DESCRIPTION

(17) The description of the embodiments of the present disclosure is intended to be illustrative and not restrictive. The embodiments disclosed in the following may be combined or substituted with each other in an advantageous situation, and other embodiments may be added to an embodiment without further description or explanation.

(18) In the following description, numerous specific details will be described in detail in order to enable the reader to fully understand the following embodiments. However, embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known structures and devices are only schematically illustrated in the drawings in order to simplify the drawings.

(19) A purpose of the present disclosure is to provide a pixel array package structure. FIG. 1A is a microscopic top view of a pixel array package structure according to one embodiment of the present disclosure. FIG. 1B is a partially enlarged schematic view of a region R1 of FIG. 1A. FIG. 1C and FIG. 1D are schematic cross-sectional views taken along line A-A' and line B-B' of FIG. 1B, respectively.

(20) Referring to FIG. 1B, FIG. 1C and FIG. 1D simultaneously, the pixel array package structure **10** includes a substrate **110**, a pixel array **120**, a reflective layer **130**, a light-absorbing layer **140**, and a light-transmitting layer **150**. The pixel array **120** is disposed on the substrate **110**.

Specifically, the pixel array **120** includes a plurality of light emitting diode (LED) chips **122**, **124**, and **126**. In various embodiments of the present disclosure, the substrate **110** may be a white substrate or a black substrate. In some embodiments, the substrate **110** may include a heat dissipation substrate, a conductive substrate, or an insulating substrate. In some embodiments, the substrate **110** has a circuit layer thereon, which is configured to electrically connect the LED chips **122**, **124**, and **126**. These LED chips include at least one red LED chip **122**, at least one green LED chip **124**, at least one blue LED chip **126**, and a combination thereof.

(21) It can be understood that the red LED chip **122** emits red light. In some other embodiments, the red LED chip **122** may also include emitting red light by exciting a wavelength conversion layer (not shown) by blue light emitted by the blue LED chip **126**. In various examples, the wavelength conversion layer can include one or more red phosphors, red quantum dots, or a mixture thereof. For example, the red phosphor material may include nitrides, fluorides and/or manganese (Mn.sup.4+) doped fluoride phosphors, such as A.sub.2[MF.sub.6]:Mn.sup.4+, in which A is Li.sup.+, Na.sup.+, K.sup.+, Rb.sup.+, Cs.sup.+ and/or NH₄.sup.+, and M is Ge, Si, Sn, Ti or Zr.

(22) It can be understood that the green LED chip **124** emits green light. In some other embodiments, the green LED chip **124** may include emitting green light by exciting a wavelength conversion layer (not shown) by blue light emitted by the blue LED chip **126**. In various examples, the wavelength conversion layer may include one or more green phosphors, green quantum dots, or

a mixture thereof. For example, the green phosphor may be silicate, LuYAG, and/or B—SiAlON, and the green quantum dots may be CdSe, CdS, CdTe, SinP, InN, AlInN, InGaInN, AlGaInN, and/or CuInGaSe. For another example, the green quantum dots may be green all-inorganic perovskite quantum dots having a chemical formula of $\text{CsPb}(\text{Br.sub.1-bI.sub.b}).\text{sub.3}$ and $0 \leq b < 0.5$.

(23) The present disclosure utilizes these LED chips **122**, **124**, and **126** with different combinations to mix the three primary colors of light (red, green, and blue) to emit white light. For example, as shown in FIG. **1B**, the pixel array package structure **10** includes a red LED chip **122**, a green LED chip **124**, and a blue LED wafer **126** arranged in a matrix with 3 columns and 1 row. In addition, the pixel array package structure **10** may also include a red LED chip **122**, a green LED chip **124**, and a blue LED chip **126** arranged as shown in FIG. **2**.

(24) For another example, as shown in FIG. **3**, the pixel array package structure **10** includes two red LED chips **122**, a green LED chip **124**, and a blue LED chip **126** arranged in a matrix with 2 columns and 2 rows. This design can increase color gamut (NTSC or BT. 2020 (Rec. 2020)) of the pixel array package structure **10**. In an embodiment, the two red LED chips **122** may respectively have different wavelength bands for better red light color rendering effect. In another embodiment, the two red LED chips **122** may have the same wavelength band but have different brightness levels for better modulation in brightness.

(25) For another example, as shown in FIG. **4**, the pixel array package structure **10** includes a red LED chip **122**, two green LED chips **124**, and a blue LED chip **126** arranged in a matrix with 2 columns and 2 rows. This design can increase the overall brightness of the pixel array package structure **10**.

(26) For example, as shown in FIG. **5**, the pixel array package structure **10** includes a red LED chip **122**, a green LED chip **124**, a blue LED chip **126**, and a white LED chip **128** arranged in a matrix with 2 columns and 2 rows. Since there is a white sub-pixel added in this design, display brightness can be improved when a backlight module with a same power is used. In other words, if a same brightness output is required, the power consumption of this design will be lower.

(27) In general, if the LEDs of the LED chips **122**, **124**, and **126** have a grain size of about 100 μm to 300 μm (for example, 150 μm), those may be called as sub-millimeter LEDs (mini LEDs). If the LEDs of the LED chips **122**, **124**, and **126** have a grain size of less than 100 μm , those may be called as micro LEDs. In application, the sub-millimeter LEDs may be applied to products such as mobile phones, televisions, car panels, and esports notebooks, and the micro LEDs may be applied to fields of wearable watches, mobile phones, car displays, augmented reality/virtual reality, display screens and televisions.

(28) Returning to FIG. **1C** and FIG. **1D**, the reflective layer **130** is disposed on the substrate **110** and between any two adjacent of the LED chips **122**, **124**, and **126**. In general, the illumination angle of each of the LED chips **122**, **124** or **126** is about 120 to 150 degrees. In order to prevent light emitted by the LED chips **122**, **124** and **126** from interfering with each other or being absorbed, the reflective layer **130** of the present disclosure is disposed between any two adjacent of the LED chips. Such design can further enhance luminous efficiency of the pixel array package structure **10**. In some embodiments, the reflective layer **130** is composed of a mixture of a colloidal material and inorganic particles. For example, the inorganic particles may include titanium dioxide (TiO.sub.2), boron nitride (BN), silicon oxide (SiO.sub.2), barium sulfate (BaSO.sub.4) or aluminum oxide (Al.sub.2O.sub.3). The colloidal material may include polymethyl methacrylate (PMMA), polycarbonate (PC), allyl diglycol carbonate (CR-39), polystyrene (PS), epoxy (epoxy), polyamide, acrylate or silicone. In one embodiment, the inorganic particles have an average particle size (D50) less than 20 μm .

(29) It should be noted that light reflectance of the reflective layer **130** depends on the concentration and characteristics of the inorganic particles in the colloidal material. The higher the concentration of the inorganic particles, the higher the light reflectance of the reflective layer **130**. The higher the light reflectance of the inorganic particles themselves, the higher the light

reflectance of the reflective layer **130**. In some embodiments of the present disclosure, the light reflectance of the reflective layer **130** should be at least greater than light absorption thereof. In other words, the light energy of the light reflected by the reflective layer **130** should be higher than the light energy of the light absorbed by the reflective layer **130**, so as to help improve the luminous efficiency of the pixel array package structure **10** and avoid rise of temperature of the reflective layer **130** due to absorption of the light energy, which will result in attenuation of the luminous efficiency of the pixel array package structure **10** due to the temperature effect. In addition, in some preferable embodiments, the light reflectance of the reflective layer **130** should be greater than about 90% to reflect the light emitted by the LED chip **122**, **124** or **126** as much as possible, and to prevent the light from being absorbed by the reflective layer **130**, which will reduce the luminous efficiency of the pixel array package structure **10**.

(30) As shown in FIGS. **1C** and **1D**, in some embodiments, the upper surface **130a** of the reflective layer **130** is substantially coplanar with the upper surface of the pixel array **120**. In various embodiments, the reflective layer **130** may be formed on the substrate **110** and filled between any two adjacent of the LED chips **122**, **124** and **126** by molding, glue-filling or other suitable processes.

(31) Referring to FIG. **1B**, FIG. **1C**, and FIG. **1D** simultaneously, the light-absorbing layer **140** is disposed on the reflective layer **130** and surrounds the pixel array **120**. In an embodiment, the lower surface **140b** of the light-absorbing layer **140** is substantially coplanar with the upper surface **120a** of the pixel array **120**. In some embodiments, the light-absorbing layer **140** may cover a portion of the reflective layer **130** located on the periphery of the pixel array **120**. In other embodiments, the light-absorbing layer **140** may also fully cover the reflective layer **130** located on the periphery of the pixel array **120**. For example, the light-absorbing layer **140** is composed of a colloidal material and an inorganic material. For example, the colloidal material may include polymethyl methacrylate (PMMA), polycarbonate (PC), allyl diglycol carbonate (CR-39), polystyrene (PS), epoxy (epoxy), polyamide, acrylate or silicone. In addition, in some embodiments, the inorganic material may be carbon powder or perovskite or the like. In a plurality of examples, the carbon powder has a specific surface area of 50 m.²/g to 70 m.²/g, such as 52 m.²/g, 54 m.²/g, 56 m.²/g, 58 m.²/g, 60 m.²/g, 62 m.²/g, 64 m.²/g, 66 m.²/g or 68 m.²/g. In another example, the light-absorbing layer **140** may also be composed of a colloidal material and an organic material. For example, the organic material may include black pigment-added or dye-added polyimide resin, poly-vinyl alcohol resin, and/or acrylate resin. In various embodiments, the light-absorbing layer **140** may form a black frame surrounding the pixel array **120** by spin coating and lithography etching or by screen printing.

(32) Referring to FIGS. **1C** and **1D**, the light-transmitting layer **150** is disposed on the pixel array **120**, the reflective layer **130**, and the light-absorbing layer **140**. It should be noted that the light-transmitting layer **150** has an upper surface **150a** and a lower surface **150b** opposite thereto, in which the lower surface **150b** is in contact with the pixel array **120**, and the upper surface **150a** has a roughness of 0.005 mm to 0.1 mm. In some examples, the light-transmitting layer **150** may include silicone or a resin. In various embodiments, the light-transmitting layer **150** may be formed by molding, glue-filling, or other suitable processes. In addition, a roughening process such as sand blasting or surface etching may be performed on the upper surface **150a** of the light-transmitting layer **150**. In various examples, the roughness of the upper surface **150a** of the light-transmitting layer **150** may be 0.01 mm, 0.02 mm, 0.03 mm, 0.04 mm, 0.05 mm, 0.06 mm, 0.07 mm, 0.08 mm, or 0.09 mm. It is assumed that the upper surface **150a** of the light-transmitting layer **150** is a smooth surface without being roughened, when the pixel array package structure **10** is illuminated by an external light source (e.g., sunlight, spot light or a light source), the viewer will not see the display screen and/or the glare problem is generated.

(33) Continuously referring to FIG. **1C** and FIG. **1D**, in one embodiment, the pixel array package structure **10** further includes an anti-external light material **160** disposed on the light-transmitting

layer **150**. In some embodiments, the anti-external light material **160** may be a polarizer or an anti-reflective coating or the like. In one example, the anti-external light material **160** may be a circular polarizer.

(34) In order to facilitate the comparison with the above embodiments and simplify the description, the same reference numerals are used to designate the same elements in the following embodiments, and the differences are mainly described for the respective embodiments, and repeated portions are not described.

(35) FIGS. **6A** and **6B** are cross-sectional views of a pixel array package structure **10a** according to another embodiment of the present disclosure, in which FIG. **6A** is a schematic cross-sectional view taken along line A-A' of FIG. **1B**, and FIG. **6B** a schematic cross-sectional view taken along line B-B' of FIG. **1B**. As shown in FIGS. **6A** and **6B**, the difference between the pixel array package structure **10a** illustrated in FIGS. **6A** and **6B** and the pixel array package structure **10** illustrated in FIGS. **1C** and **1D** is that the light-absorbing layer **140** of the pixel array package structure **10a** is embedded in the reflective layer **130**. In an embodiment, the upper surface **140a** of the light-absorbing layer **140** is substantially coplanar with the upper surface **120a** of the pixel array **120**. Specifically, in the embodiment, the reflective layer **130** has a first portion **130P.sub.1**, a second portion **130P.sub.2**, and a third portion **130P.sub.3**, in which the first portion **130P.sub.1** is located on the periphery of the pixel array **120** and interposed between the substrate **110** and the light-absorbing layer **140**; the second portion **130P.sub.2** is an upward extension of the first portion **130P.sub.1**, which surrounds the pixel array **120** and is interposed between the pixel array **120** and the light-absorbing layer **140**; and the third portion **130P.sub.3** is located between any two adjacent of the LED chips **122**, **124** and **126** of the pixel array **120**. It should be noted that in other embodiments, the reflective layer **130** may not have the second portion **130P.sub.2**, that is, the boundary of the light-absorbing layer **140** may be in close proximity to the periphery of the pixel array **120**.

(36) FIGS. **7A** and **7B** are cross-sectional views of a pixel array package structure **10b** according to another embodiment of the present disclosure, in which FIG. **7A** is a schematic cross-sectional view taken along line A-A' of FIG. **1B**, and FIG. **7B** is a schematic cross-sectional view taken along line B-B' of FIG. **1B**. As shown in FIGS. **7A** and **7B**, the difference between the pixel array package structure **10b** illustrated in FIGS. **7A** and **7B** and the pixel array package structure **10** illustrated in FIGS. **1C** and **1D** is that the upper surface **140a** of the light-absorbing layer **140** of the pixel array package structure **10b** is substantially coplanar with the upper surface **150a** of the light-transmitting layer **150**.

(37) In an embodiment, the lower surface **140b** of the light-absorbing layer **140** is substantially coplanar with the upper surface **120a** of the pixel array **120**. In an embodiment, the light-transmitting layer **150** may cover a portion of the reflective layer **130**, that is, the light-absorbing layer **140** only covers a portion of the reflective layer **130** located at the periphery of the pixel array **120**. In other embodiments, the light-absorbing layer **140** fully covers the reflective layer **130** located at the periphery of the pixel array **120**.

(38) FIG. **8** is a top view of a pixel array package structure according to a variant embodiment of the present disclosure. As shown in FIG. **8**, the difference between the pixel array package structure **20** illustrated in FIG. **8** and the pixel array package structure **10** illustrated in FIG. **1B** is that the pixel array package structure **20** may include a first group pixel array **120S.sub.1** and a second group pixel array **120S.sub.2**, in which each of the first group pixel array **120S.sub.1** and the second group pixel array **120S.sub.2** includes a red LED chip **122**, a green LED chip **124** and a blue LED chip **126**. The display application usually includes thousands or tens of thousands of pixel array package structures, and when one of a certain group LED chips of a certain pixel array package structure does not illuminate, another group LED chips will continue to operate by such design. It does not have a visual impact on the user and can reduce maintenance costs.

(39) FIG. **9A** is a microscopic top view of a pixel array package structure **30** according to a variant

embodiment of the present disclosure. FIG. 9B is a partially enlarged view of a region R2 of FIG. 9A. FIG. 9C is a schematic cross-sectional view taken along line A-A' of FIG. 9B. As shown in FIGS. 9B and 9C, the pixel array package structure 30 includes a substrate 110, a pixel array 120, a light-absorbing layer 140, and a light-transmitting layer 150. Specifically, the pixel array 120 is disposed on the substrate 110. The pixel array 120 includes a plurality of LED chips, such as at least one red LED chip 122, at least one green LED chip 124, at least one blue LED chip 126, and a combination thereof. The light-absorbing layer 140 is disposed on the substrate 110 and surrounds the pixel array 120. In the present embodiment, the upper surface of the light-absorbing layer 140 is substantially coplanar with the upper surface 120a of the pixel array 120. The light-transmitting layer 150 is disposed on the light-absorbing layer 140 and the pixel array 120. The light-transmitting layer 150 has an upper surface 150a and a lower surface 150b opposite thereto. The lower surface 150b of the light-transmitting layer 150 is in contact with the pixel array 120 and the light-absorbing layer 140, and the upper surface 150a thereof has a roughness of 0.005 mm to 0.1 mm. In one embodiment, the pixel array package structure 30 may further include an anti-external light material (not shown) disposed on the upper surface 150a of the light-transmitting layer 150.

(40) FIG. 10A is a top view of a pixel array package structure 40 according to a variant embodiment of the present disclosure. FIG. 10B is a schematic cross-sectional view taken along line A-A' of FIG. 10A. As shown in FIGS. 10A and 10B, the pixel array package structure 40 includes a substrate 110, a pixel array 120, a light-absorbing layer 140, and a light-transmitting layer 150. Specifically, the pixel array 120 is disposed on the substrate 110. The pixel array 120 includes a plurality of LED chips, such as at least one red LED chip 122, at least one green LED chip 124, at least one blue LED chip 126, and a combination thereof. The light-absorbing layer 140 is disposed at both sides of the pixel array 120. The light-transmitting layer 150 is disposed between the light-absorbing layers 140 and surrounds the pixel array 120. In more detail, the light-transmitting layer 150 has a first portion 150P.sub.1 and a second portion 150P.sub.2, in which the first portion 150P.sub.1 is disposed on the pixel array 120, and the second portion 150P.sub.2 is disposed beneath the first portion 150P.sub.1 and interposed between the pixel array 120 and the light-absorbing layer 140. In the present embodiment, a top surface 150T of the light-transmitting layer 150 is substantially coplanar with the top surface 140T of the light-absorbing layer 140. In one embodiment, the pixel array package structure 40 further includes an anti-external light material (not shown) disposed on the top surface 150T of the light-transmitting layer 150 and the top surface 140T of the light-absorbing layer 140.

(41) FIG. 11 is a graph comparing brightness of pixel array structures of the present disclosure with a conventional pixel array structure. As shown in FIG. 11, A represents a pixel array package structure 10, 10a or 10b of the present disclosure; B represents a pixel array package structure 30 of the present disclosure; C represents a pixel array package structure 40 of the present disclosure; and P represents a conventional pixel array structure. It should be noted that a light absorbing particles-added light-transmitting layer is used to encapsulate a pixel array of the conventional pixel array structure P. As shown in FIG. 11, the vertical axis represents brightness measured when a pixel array package structure is illuminated, and the horizontal axis represents a brightness ratio of an unilluminated pixel array package structure adjacent to the illuminated pixel array package structure. That is, the greater the brightness of the vertical axis, the greater the luminous intensity of the pixel array package structure. The smaller the brightness ratio of the horizontal axis, the lower the effect on the unilluminated pixel array package structure from the illuminated pixel array package structure. The brightness ratio of the pixel array package structure of the present disclosure is preferably less than 0.15%. As can be seen from FIG. 11, the brightness of the conventional pixel array structure P is about 150 cd/m.sup.2, and the brightness ratio thereof is about 0.267%. The brightness of the pixel array structure A of the present disclosure is about 310 cd/m.sup.2, and the brightness ratio thereof is about 0.14%. The brightness of the pixel array structure B of the present disclosure is about 97 cd/m.sup.2, and the brightness

ratio thereof is about 0.036%. The brightness of the pixel array structure C of the present disclosure is about 105 cd/m.sup.2, and the brightness ratio thereof is about 0.013%. Although the brightness ratio of the pixel array structures B and C of the present disclosure is much smaller than the brightness ratio of the pixel array structure A of the present disclosure, the brightness of the pixel array structures B and C is relatively low, and therefore, the pixel array structure A of the present disclosure is a preferable embodiment. In addition, compared the pixel array structure A with the conventional pixel array structure P, the brightness and brightness ratio of the pixel array structure A are better than those of the conventional pixel array structure P. Therefore, the pixel array structure A of the present disclosure not only can solve the problem of reduced light output of the forward light of the conventional pixel array structure P, but also can solve the problem of poor contrast.

(42) FIG. 12 is a top view of a display panel 200 according to one embodiment of the present disclosure. FIG. 13 is a partially enlarged view of FIG. 12. Referring to FIG. 12 first, the display panel 200 includes a plurality of sub-display panels 210. Specifically, any two adjacent of the sub-display panels 210 have a splicing gap D1 therebetween. In various embodiments, the splicing gap D1 between the sub-display panels 210 is less than 100 μm. For the viewer, the presence of these splicing gaps D1 may cause the display screen to be cut and visually discontinuous.

(43) Referring to FIG. 13, each of the sub-display panels 210 includes a plurality of pixel array package structures 10 as described above. In more detail, in various embodiments, any two adjacent of the pixel array package structures 10 have a pitch D2 therebetween. When the LED chips in the pixel array package structure 10 are designed as micro LEDs, the pitch D2 between the pixel array package structures 10 is less than about 0.5 mm. When the pixel array package structures 10 are designed as mini LEDs, the pitch D2 between the pixel array package structures 10 is about 0.5 mm to 1.0 mm. In the embodiment of the mini LEDs, the width W of the light-absorbing layer 140 in the pixel array package structure 10 is in a range of 50 μm to 650 μm, such as 100 μm, 150 μm, 200 μm, 250 μm, 300 μm, 350 μm, 400 μm, 450 μm, 500 μm, 550 μm, or 600 μm. A single sub-display panel 210 may include tens of thousands to hundreds of thousands of the pixel array package structures 10, and any two adjacent of the pixel array package structures 10 have a sub-splicing gap (2W+D2). In the case where the single sub-display panel 210 has been divided by a plurality of sub-splicing gaps (2W+D2), the splicing gaps D1 between these sub-display panels 210 (indicated in FIG. 12) is not obvious to the viewer.

(44) In other embodiments, each of the sub-display panels 210 may also include a plurality of the pixel array package structures 10a, 10b, 20, 30 or 40 as described above.

(45) In summary, the various embodiments provided by the present disclosure can increase the brightness of the LED chips without losing the contrast. Since the position of lateral light output of the LED chip is encapsulated by an opaque material or a highly reflective material, and the position of forward light output of the LED chip is encapsulated by a high transmittance material, the problem of low light efficiency resulting from absorption of the light emitted by the LED by the light-absorbing particles in the prior art can be improved. In addition, the sub-splicing gaps in the single sub-display panel constituted by the light-absorbing layer and the pitch of any two adjacent of the pixel array package structures can make the viewer feel seamlessly splicing visual effects of the display panel.

(46) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A pixel array package structure, comprising: a substrate; a pixel array disposed on the substrate, wherein the pixel array comprises a plurality of light emitting diode chips, and the plurality of light emitting diode chips comprise a red diode chip, a green diode chip, a blue diode chip, and a combination thereof; a reflective layer disposed on the substrate and between any two adjacent of the plurality of light emitting diode chips; a light-absorbing layer embedded in the reflective layer and surrounding the pixel array; and a light-transmitting layer disposed on the pixel array, the reflective layer, and the light-absorbing layer, wherein the light-transmitting layer has an upper surface and a lower surface opposite thereto, and the lower surface is in contact with the pixel array.
2. The pixel array package structure of claim 1, further comprising an anti-external light material disposed on the light-transmitting layer.
3. The pixel array package structure of claim 2, wherein the anti-external light material is a polarizer.
4. The pixel array package structure of claim 1, wherein the reflective layer is composed of a mixture of a colloidal material and inorganic particles, wherein the inorganic particles comprise titanium dioxide, boron nitride, silicon dioxide, barium sulfate or aluminum oxide.
5. The pixel array package structure of claim 1, wherein an upper surface of the light-absorbing layer is substantially coplanar with an upper surface of the pixel array.
6. The pixel array package structure of claim 1, wherein the light-absorbing layer is composed of a colloidal material and an inorganic material, or a colloidal material and an organic material.
7. The pixel array package structure of claim 6, wherein the inorganic material is carbon powder or perovskite.
8. The pixel array package structure of claim 7, wherein the carbon powder has a specific surface area of 50 m²/g to 70 m²/g.
9. A display panel, comprising: a plurality of sub-display panels, any two adjacent of the sub-display panels having a splicing gap, wherein each of the sub-display panels comprises a plurality of the pixel array package structures of claim 1.
10. The display panel of claim 9, further comprising an anti-external light material disposed on the light-transmitting layer.
11. The display panel of claim 10, wherein the anti-external light material is a polarizer.
12. The display panel of claim 9, wherein the reflective layer is composed of a mixture of a colloidal material and inorganic particles, wherein the inorganic particles comprise titanium dioxide, boron nitride, silicon dioxide, barium sulfate or aluminum oxide.
13. The display panel of claim 9, wherein an upper surface of the light-absorbing layer is substantially coplanar with an upper surface of the pixel array.
14. The display panel of claim 9, wherein the light-absorbing layer is composed of a colloidal material and an inorganic material, or a colloidal material and an organic material.
15. The display panel of claim 14, wherein the inorganic material is carbon powder or perovskite.
16. The display panel of claim 15, wherein the carbon powder has a specific surface area of 50 m²/g to 70 m²/g.
17. The display panel of claim 9, wherein the splicing gap between the sub-display panels is less than 100 μ m.
18. The display panel of claim 9, wherein any two adjacent of the pixel array package structures have a pitch therebetween.
19. The display panel of claim 18, wherein the LED chips in the pixel array package structure are micro LEDs, the pitch between the pixel array package structures is less than 0.5 mm.

20. The display panel of claim 18, wherein the LED chips in the pixel array package structure are mini LEDs, the pitch between the pixel array package structures is about 0.5 mm to 1.0 mm.
